

Section 13.4 covers parameter estimation using Kalman filtering. Sometimes, an engineer wants to estimate the parameters of a system but does not care about estimating the states. This becomes a system identification problem. The system equations are generally nonlinear functions of the system parameters. System parameters are usually considered to be constant, or slowly time-varying, and a nonlinear Kalman filter (or any other nonlinear state estimator) can be adapted to estimate system parameters.

13.1 THE LINEARIZED KALMAN FILTER

In this section, we will show how to linearize a nonlinear system, and then use Kalman filtering theory to estimate the deviations of the state from a nominal state value. This will then give us an estimate of the state of the nonlinear system. We will derive the linearized Kalman filter from the continuous-time viewpoint, but the analogous derivation for discrete-time or hybrid systems are straightforward.

Consider the following general nonlinear system model:

$$\begin{aligned}\dot{x} &= f(x, u, w, t) \\ y &= h(x, v, t) \\ w &\sim (0, Q) \\ v &\sim (0, R)\end{aligned}\tag{13.1}$$

The system equation $f(\cdot)$ and the measurement equation $h(\cdot)$ are nonlinear functions. We will use Taylor series to expand these equations around a nominal control u_0 , nominal state x_0 , nominal output y_0 , and nominal noise values w_0 and v_0 . These nominal values (all of which are functions of time) are based on *a priori* guesses of what the system trajectory might look like. For example, if the system equations represent the dynamics of an airplane, then the nominal control, state, and output might be the planned flight trajectory. The *actual* flight trajectory will differ from this nominal trajectory due to mismodeling, disturbances, and other unforeseen effects. But the actual trajectory should be close to the nominal trajectory, in which case the Taylor series linearization should be approximately correct. The Taylor series linearization of Equation (13.1) gives

$$\begin{aligned}\dot{x} &\approx f(x_0, u_0, w_0, t) + \left. \frac{\partial f}{\partial x} \right|_0 (x - x_0) + \left. \frac{\partial f}{\partial u} \right|_0 (u - u_0) + \\ &\quad \left. \frac{\partial f}{\partial w} \right|_0 (w - w_0) \\ &= f(x_0, u_0, w_0, t) + A\Delta x + B\Delta u + L\Delta w \\ y &\approx h(x_0, v_0, t) + \left. \frac{\partial h}{\partial x} \right|_0 (x - x_0) + \left. \frac{\partial h}{\partial v} \right|_0 (v - v_0) \\ &= h(x_0, v_0, t) + C\Delta x + M\Delta v\end{aligned}\tag{13.2}$$

The definitions of the partial derivative matrices A , B , C , L , and M are apparent from the above equations. The 0 subscript on the partial derivatives means that they are evaluated at the nominal control, state, output, and noise values. The definitions of the deviations Δx , Δu , Δw , and Δv are also apparent from the above equations.

Let us assume that the nominal noise values $w_0(t)$ and $v_0(t)$ are both equal to 0 for all time. [If they are not equal to 0 then we should be able to write them as the sum of a known deterministic part and a zero-mean part, redefine the noise quantities, and rewrite Equation (13.1) so that the nominal noise values are equal to 0. See Problem 13.1]. Since $w_0(t)$ and $v_0(t)$ are both equal to 0, we see that $\Delta w(t) = w(t)$ and $\Delta v(t) = v(t)$. Further assume that the control $u(t)$ is perfectly known. In general, this is a reasonable assumption. After all, the control input $u(t)$ is determined by our control system, so there should not be any uncertainty in its value. This means that $u_0(t) = u(t)$ and $\Delta u(t) = 0$. However, in reality there may be uncertainties in the outputs of our control system because they are connected to actuators that have biases and noise. If this is the case then we can express the control as $u_0(t) + \Delta u(t)$, where $u_0(t)$ is known and $\Delta u(t)$ is a zero-mean random variable, rewrite the system equations with a perfectly known control signal, and include $\Delta u(t)$ as part of the process noise (see Problem 13.2). Now we define the nominal system trajectory as

$$\begin{aligned}\dot{x}_0 &= f(x_0, u_0, w_0, t) \\ y_0 &= h(x_0, v_0, t)\end{aligned}\tag{13.3}$$

We define the deviation of the true state derivative from the nominal state derivative, and the deviation of the true measurement from the nominal measurement, as follows:

$$\begin{aligned}\Delta \dot{x} &= \dot{x} - \dot{x}_0 \\ \Delta y &= y - y_0\end{aligned}\tag{13.4}$$

With these definitions Equation (13.2) becomes

$$\begin{aligned}\Delta \dot{x} &= A\Delta x + Lw \\ &= A\Delta x + \tilde{w} \\ \tilde{w} &\sim (0, \tilde{Q}), \quad \tilde{Q} = LQL^T \\ \Delta y &= C\Delta x + Mv \\ &= C\Delta x + \tilde{v} \\ \tilde{v} &\sim (0, \tilde{R}), \quad \tilde{R} = MRM^T\end{aligned}\tag{13.5}$$

The above equation is a linear system with state Δx and measurement Δy , so we can use a Kalman filter to estimate Δx . The inputs to the filter consist of Δy , which is the difference between the actual measurement y and the nominal measurement y_0 . The Δx that is output from the Kalman filter is an estimate of the difference between the actual state x and the nominal state x_0 . The Kalman filter equations for the linearized Kalman filter are

$$\begin{aligned}\Delta \hat{x}(0) &= 0 \\ P(0) &= E[(\Delta x(0) - \Delta \hat{x}(0))(\Delta x(0) - \Delta \hat{x}(0))^T] \\ \Delta \dot{\hat{x}} &= A\Delta \hat{x} + K(\Delta y - C\Delta \hat{x}) \\ K &= PC^T \tilde{R}^{-1} \\ \dot{P} &= AP + PA^T + \tilde{Q} - PC^T \tilde{R}^{-1} CP \\ \hat{x} &= x_0 + \Delta \hat{x}\end{aligned}\tag{13.6}$$

For the Kalman filter, P is equal to the covariance of the estimation error. In the linearized Kalman filter this is no longer true because of errors that creep into the linearization of Equation (13.2). However, if the linearization errors are small then P should be approximately equal to the covariance of the estimation error. The linearized Kalman filter can be summarized as follows.

The continuous-time linearized Kalman filter

1. The system equations are given as

$$\begin{aligned} \dot{x} &= f(x, u, w, t) \\ y &= h(x, v, t) \\ w &\sim (0, Q) \\ v &\sim (0, R) \end{aligned} \quad (13.7)$$

The nominal trajectory is known ahead of time:

$$\begin{aligned} x_0 &= f(x_0, u_0, 0, t) \\ y_0 &= h(x_0, 0, t) \end{aligned} \quad (13.8)$$

2. Compute the following partial derivative matrices evaluated at the nominal trajectory values:

$$\begin{aligned} A &= \left. \frac{\partial f}{\partial x} \right|_0 \\ L &= \left. \frac{\partial f}{\partial w} \right|_0 \\ C &= \left. \frac{\partial h}{\partial x} \right|_0 \\ M &= \left. \frac{\partial h}{\partial v} \right|_0 \end{aligned} \quad (13.9)$$

3. Compute the following matrices:

$$\begin{aligned} \tilde{Q} &= LQL^T \\ \tilde{R} &= MRM^T \end{aligned} \quad (13.10)$$

4. Define Δy as the difference between the actual measurement y and the nominal measurement y_0 :

$$\Delta y = y - y_0 \quad (13.11)$$

5. Execute the following Kalman filter equations:

$$\begin{aligned} \Delta \hat{x}(0) &= 0 \\ P(0) &= E[(\Delta x(0) - \Delta \hat{x}(0))(\Delta x(0) - \Delta \hat{x}(0))^T] \\ \Delta \dot{\hat{x}} &= A\Delta \hat{x} + K(\Delta y - C\Delta \hat{x}) \\ K &= PC^T \tilde{R}^{-1} \\ \dot{P} &= AP + PA^T + \tilde{Q} - PC^T \tilde{R}^{-1} CP \end{aligned} \quad (13.12)$$